

Day 2 — Cloud Architecture & Infrastructure

Activity packet for participant triads. Today's job: take the entities from Day 1 and the components from the TCD, decide where they **physically run**, and draft TMD Section 2 — Cloud topology.

Where we are in the week

Day 1 produced the entity model. Today decides **where each entity and each component runs in the cloud** — region, availability zone, managed service vs custom, multi-tenancy stance, and the cost / latency trade-offs that fall out.

By 16:00, every triad has TMD Section 2 with a topology that an SRE / platform engineer would accept.

Inputs

- TMD Section 1 (data model)
 - TCD Section 2 (Container diagram)
 - TCD Section 4 (SLOs — region/AZ choices affect availability and latency budgets)
 - TCD Section 3 (compliance — data residency drives region choice)
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The cloud vocabulary card (today's reference)

Term	What it means	TPM-relevant signal
Region	A geographic cloud location (us-east-1, eu-west-2)	Drives data residency + base latency to users
Availability Zone (AZ)	A failure-isolated facility within a region	Multi-AZ = survives single facility failure
Multi-region	Deployment across regions	Survives whole region failure; expensive
Managed service	Cloud provider runs it (RDS, S3, SQS)	Less ops cost; less control
Self-managed	You run it on raw compute	More control; more ops cost
Multi-tenant	One deployment serves all customers	Lower cost; harder isolation
Single-tenant	Each customer gets their own deployment	Stronger isolation; higher cost
Edge / CDN	Static and cached content served closest to users	Reduces latency for static assets

The five cloud-topology decisions

For most features, the topology decision reduces to five choices. Today's TMD Section 2 documents each:

1. **Region choice** — which region(s)?
2. **Multi-AZ stance** — single AZ, multi-AZ, multi-region?
3. **Managed vs self-managed** — for each component, who runs it?
4. **Multi-tenancy stance** — shared / dedicated / hybrid?
5. **Network boundary** — public, private, VPN, direct connect?

Each decision has a **cost dimension** and a **risk dimension**. The TPM job: surface both, name the trade-off, and frame the choice in customer terms.

Activity 1 — Region & Availability Zone Choice

Format: Triad • 35 min • Block 1

Purpose

Decide which region(s) the feature deploys to and which multi-AZ stance it takes.
Defend in customer + compliance + cost terms.

Setup

Each triad needs PRD Section 2 (customer base), TCD Section 3 (compliance), TCD Section 4 (SLOs), and the Week-4 trade-off template. AI optional.

Triad protocol

1. **Pull customer-location facts from PRD Section 2** (5 min). Where do users live? Where is the bulk of traffic?
2. **Pull compliance facts from TCD Section 3** (5 min). Any data-residency requirements?
3. **Decide on region(s)** (10 min). Single region? Multi-region active-passive? Multi-region active-active?
4. **Decide on multi-AZ** (10 min). At minimum, 2-AZ for any production workload. 3-AZ for stateful workloads with quorum requirements.
5. **Capture the trade-off** (5 min). Use the Week-4 template.

Worked example — FieldPulse reconcile

Region + AZ stance

```
**Region:** us-east-1 (primary), us-west-2 (warm-DR).  
- Customer base: 92% US-based; 8% Canada (cross-region acceptable  
  for Canadian users; latency ~80ms additional).  
- Compliance: SOC 2 audit retention requires named region; no  
  EU residency requirement currently.  
  
**AZ stance:** Multi-AZ active-active for all stateful tiers  
  (Postgres primary + 2 read replicas across 3 AZs).  
  
**Trade-off — multi-region active-active vs warm-DR:**  
**Option A:** us-east-1 primary, us-west-2 warm-DR (RPO 5 min,  
  RTO 30 min).  
**Option B:** Both regions live, traffic split via DNS.  
**Choice:** Option A.  
**Why:** Active-active doubles cost and adds cross-region  
  consistency complexity; we don't have evidence of single-  
  region outages costing more than the doubled cost. Multi-AZ  
  within us-east-1 satisfies our 99.5% availability SLO.  
**Accepted cost:** A full us-east-1 outage would breach SLO  
  for ~30 minutes.  
**Revisit trigger:** A second us-east-1 multi-hour outage in 12  
  months, OR availability SLO tightens to 99.9%+.
```

Output

A regional-stance section ready for TMD Section 2.

Activity 2 — Managed vs Self-Managed for Each Component

Format: Triad • 40 min • Block 2

Purpose

For each container in the TCD's Container diagram, decide whether to use a managed service or self-manage. Defend each.

Setup

Each triad needs the TCD Container diagram and the four-question test card. AI optional.

The "managed service first" default

Default: **use the managed service unless you have a specific reason not to.**

Reasons to self-manage:

- The managed service's failure modes don't match your needs
- The managed service is significantly more expensive at your scale
- Compliance requires data control the provider can't offer
- Your team has unusual operational expertise that goes to waste

For most B2B features, the answer is **managed**. Don't reinvent.

Triad protocol

1. **List each container** (5 min). Pull from TCD Section 2.
2. **For each, identify the managed-service option** (10 min). e.g., RDS for Postgres, MSK for Kafka, ElastiCache for Redis, ALB for the load balancer.
3. **For each, decide managed or self** (15 min). Use the four-question test:
 - Do failure modes match our needs?
 - Is cost reasonable at expected scale?
 - Does compliance allow it?
 - Do we have a specific reason to control this?
4. **Capture the table** (10 min):

Component	Managed option	Choice	Why
Postgres	AWS RDS	Managed (RDS)	Default; team ops capacity is finite
Kafka	AWS MSK	Managed (MSK)	Same
Object storage	S3	Managed (S3)	Industry-standard durability
Custom AI service (if any)	None	Self-managed	No managed equivalent

What "good" looks like

- Most components are **managed**
- Self-managed choices have a **specific** justification — not "we want control"
- The cost dimension is named (rough — "comparable to RDS at our size")

Deliverable

Managed-vs-self table covering every TCD container, with a specific reason for each self-managed choice.

Activity 3 — Multi-Tenancy + Network Boundary

Format: Triad • 40 min • Block 3

Purpose

Decide the multi-tenancy stance and the network boundary. Both are stakeholder-touching decisions.

Setup

Each triad needs PRD Section 1 (customer context), the tenancy spectrum card, and the TCD Section 6 sign-off matrix. AI optional.

Multi-tenancy spectrum

Stance	Architecture	Pros	Cons
Shared	One deployment serves all tenants	Lowest cost; easiest ops	Noisy neighbor; weakest isolation
Pooled	One deployment, but per-tenant limits + isolation	Lower cost than dedicated; better isolation	Complex tenant-aware code
Dedicated	One deployment per tenant	Strongest isolation; per-tenant compliance	High cost; complex deployment
Hybrid	Most tenants shared; some dedicated	Match cost to value	Operational overhead of two patterns

Most B2B SaaS at FieldPulse's stage uses **pooled with per-tenant rate limits** (revisited from Week 4 Day 4).

Triad protocol

1. **Identify your customers' tenancy expectations** (10 min). What did your PRD Section 1 customer say? Are any large customers contractually expecting dedicated infra?
2. **Pick a stance** (10 min). Pooled is the default; defend if you choose dedicated.
3. **Network boundary** (15 min). Decide:
 - Public internet (default for SaaS APIs)
 - Customer-VPN-only (some enterprise contracts)
 - Direct Connect / Private Link (large enterprise, regulated industries)
4. **Stakeholder implications** (5 min). Who needs to sign off? Add to TCD Section 6 if not already there.

Worked example

Multi-tenancy stance

****Choice:**** Pooled.

****Why:**** All current and prospective customers fit the small-mid shop profile. None have contractually expected dedicated infra. Pooled with per-tenant rate limits (NFR from TCD Section 4) provides isolation against runaway scripts.

****Accepted cost:**** A noisy-neighbor incident from one tenant could affect others. Mitigated by rate limits (TCD Section 4).

****Revisit trigger:**** A customer larger than 5x our current largest signs, OR a regulated-industry customer requires dedicated.

Network boundary

****Choice:**** Public HTTPS for the API + mobile app traffic.
Internal services in a private VPC.

****Why:**** Standard for B2B SaaS at our segment. No customer has requested VPN-only.

****Revisit trigger:**** Compliance (e.g., HIPAA-bound customer) or contractual demand.

Deliverable

Tenancy stance and network boundary documented with rationale, revisit triggers, and updated sign-off matrix entries for any new stakeholders.

Activity 4 — Cost Awareness + AI Sanity Check

Format: Triad • 45 min + Wrap • Block 4

Purpose

A TMD Section 2 without a cost dimension reads as wishful thinking. Today's last block: rough out the cost shape and use AI to surface gaps.

Setup

Each triad needs the topology decisions from Activities 1–3, rough pricing references (RDS / MSK / S3 / ALB), and the AI sanity-check prompt. AI required.

Cost awareness — the rough-order-of-magnitude pass

For each component, capture a rough monthly cost band. We're not pricing exactly — we're surfacing relative scale.

Component	Stance	ROM cost / month	Cost driver
Postgres (RDS)	Multi-AZ, db.m5.large	\$400–\$700	Instance hours; storage; IOPS
Read replica	Single-AZ	\$200–\$350	Instance hours
Kafka (MSK)	3-broker, m5.large	\$700–\$1100	Brokers; storage; egress
S3 (audit)	Standard tier	\$50 / TB-month	Storage; PUT/GET requests
ALB	Standard	\$25 + traffic	Hours + LCUs
Outbound bandwidth (estimated)		\$50–\$150	\$/GB egress to internet

The numbers are illustrative — engineering will refine. The TPM job: catch the **2x cost surprise** before it ships.

AI sanity check

```
Role: SRE reviewing a feature's cloud topology.
Context: <paste TMD Section 2 sections – region, AZ, managed services,
        tenancy, network boundary, ROM cost>
Task: Identify the top 3 risks in this topology that could
      surprise the team within 6 months.
Constraints:
  - Be specific to the configurations described
  - For each: scenario, likelihood (L/M/H), suggested mitigation
  - Do not invent platform features
Format: Numbered – Risk / Scenario / Likelihood / Mitigation.
```

Triad protocol

1. **ROM cost table** (15 min). Rough numbers for each component.
2. **AI sanity check** (10 min). Run the prompt; capture the 3 risks.
3. **Adopt / defer / reject** (10 min).
4. **Update Section 2** (10 min). Polish for sharing.

Deliverable

Polished TMD Section 2 with ROM cost table, AI-surfaced topology risks adopted/deferred/rejected, and a provenance note.

Wrap (last 15 min)

Each triad shares:

- Their region + AZ stance, in one sentence
 - One trade-off they made (with revisit trigger)
 - One cost surprise they want to flag for review
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End-of-day checkpoint

Each triad ends Day 2 with:

- ✓ Region + AZ stance with trade-off
- ✓ Managed-vs-self table for every component
- ✓ Multi-tenancy stance with revisit trigger
- ✓ Network boundary stance
- ✓ **ROM cost table**
- ✓ AI provenance log entry
- ✓ TMD Section 2 drafted